

Regularized Relational Projector Geometry (RRPG)

A basis-covariant projector-curvature framework with local variational self-consistency dynamics

Research framework manuscript

Date: 2026-06-06

Abstract

Regularized Relational Projector Geometry (RRPG) is a mathematical framework in which geometry is defined by spectral projectors over a relational substrate rather than by an assumed background manifold. The primitive object is a normalized symmetric relational operator A . Local geometric structure is obtained from a rank- k spectral projector $P(A)$, and curvature is represented by a finite-difference noncommutativity obstruction Ω generated by constraint deformations of the substrate. The framework separates intrinsic basis-covariant objects, including P , projector distance, Ω , and the regularized curvature target $R_{\{\Omega, \tau\}}$, from node-basis diagnostics such as K_i , S_r , and core-enrichment statistics C_r . The central dynamics is a regularized local relaxation law $R_{\{\Omega, \tau\}}[A] = \text{Norm}(P(-\Omega^2)P + \tau A)$, together with the self-consistency functional $E_{\{\text{RRPG}, \tau\}}(A) = \|A - R_{\{\Omega, \tau\}}[A]\|_F^2$. Integrated numerical certification across matrix sizes and projector ranks supports basis covariance, basis invariance of the energy, finite-difference variational descent, and coarse projector-curvature stability above null dynamics. The current evidence supports RRPG as a local projector-curvature framework with robust coarse geometric observables, while global sector-attractor dynamics and complete holonomy/transport dynamics remain open extensions.

Keywords: relational geometry; spectral projectors; curvature obstruction; covariant dynamics; variational relaxation; emergent geometry; pre-geometric models

1. Introduction

RRPG proposes that geometric information can be extracted from relational operators by tracking the behavior of spectral projectors under constrained deformations. Instead of assuming a background metric and then studying fields on top of it, RRPG begins with a relational substrate A and defines geometry through the projector $P(A)$ selected from its spectrum. Curvature is not introduced as a tensor over a pre-existing manifold; it appears as an obstruction to commuting infinitesimal deformations of the projector.

The purpose of this paper is to give a self-contained formulation of RRPG as a basis-covariant projector-curvature framework with a regularized variational relaxation law. The framework is intentionally conservative: it defines a robust local geometry and a tested local dynamics, while explicitly separating more speculative transport and holonomy dynamics as future work.

The main contribution is the regularized curvature target $R_{\{\Omega, \tau\}}[A]$. The regularization term τA prevents the curvature-power target from becoming numerically or conceptually singular when $P(-\Omega^2)P$ is very small. This yields a stable, covariant, locally variational self-consistency dynamics.

$$R_{\{\Omega, \tau\}}[A] = \text{Norm}(P(-\Omega^2)P + \tau A) \quad (1)$$

$$E_{\{\text{RRPG}, \tau\}}(A) = \|A - R_{\{\Omega, \tau\}}[A]\|_F^2 \quad (2)$$

2. Relational substrate and projector geometry

Let A be a real symmetric normalized relational substrate. In applications, A can be interpreted as a weighted relation matrix, correlation operator, adjacency-like substrate, or abstract finite relational operator. RRPG does not require the entries of A to be coordinates in a physical space; their role is relational rather than spatial.

$$A = A^T, \quad \|A\|_F = 1 \quad (3)$$

Given a rank parameter k , the primary geometric object is the rank- k spectral projector onto a selected spectral sector of A . In the finite-dimensional formulation used here, the projector is built from the top- k eigenvectors of the constrained substrate.

$$P(A) = B_k(A) B_k(A)^T \quad (4)$$

The natural projector distance between two relational states is the Frobenius distance between projectors. This distance is intrinsic to the selected subspace and avoids treating individual node labels as invariant geometric points.

$$d_P(A, B) = \|P(A) - P(B)\|_F \quad (5)$$

3. Curvature obstruction and diagnostic sectors

To define curvature, introduce two symmetric constraint directions C_1 and C_2 . These directions generate local deformations of the relational substrate. The deformed substrate is:

$$A(a, b) = (I + \alpha(aC_1 + bC_2)) A (I + \alpha(aC_1 + bC_2)) \quad (6)$$

The projector field $P(a, b) = P(A(a, b))$ can then be differentiated by finite differences. The curvature obstruction is the projected commutator of the two projector derivatives.

$$\partial_1 P \approx [P(\delta, 0) - P(-\delta, 0)] / (2\delta) \quad (7)$$

$$\partial_2 P \approx [P(0, \delta) - P(0, -\delta)] / (2\delta) \quad (8)$$

$$\Omega_{12} = P[\partial_1 P, \partial_2 P]P \quad (9)$$

The operator Omega is a local curvature or noncommutativity obstruction. It is basis-covariant under orthogonal changes of representation and does not depend on a fixed coordinate geometry. For diagnostic purposes one may define a node-basis curvature profile K_i , but this profile is not itself a fully basis-invariant object.

$$K_i = \sqrt{\sum_j (\Omega_{12})^2_{ij}} \quad (10)$$

$$S_r = \text{top-}r \text{ indices of } 0.65 \text{ Norm}(\text{diag}(P)) + 0.35 \text{ Norm}(K) \quad (11)$$

$$C_r = [\sum_{i \in S_r} K_i / \sum_i K_i] / [r/N] \quad (12)$$

The sector sets S_r and enrichment factors C_r are useful diagnostics in a chosen relational node basis. They should be interpreted as resolution-dependent observables rather than invariant geometric primitives.

4. Regularized covariant target and self-consistency energy

The intrinsic curvature-power operator is built from the square of the curvature obstruction. Since Omega transforms covariantly, the operator $P(-\Omega^2)P$ is also covariant.

$$C_\Omega[A] = P(-\Omega^2)P \quad (13)$$

In finite systems, C_Ω can be very small. Direct normalization of a near-zero curvature-power operator can amplify numerical noise and obscure the intended dynamics. RRPG therefore uses a regularized target.

$$R_{\{\Omega, \tau\}}[A] = \text{Norm}(C_\Omega[A] + \tau A) \quad (14)$$

The parameter tau is small and positive. The certification audit supports $\tau = 10^{-5}$ for the tested finite-dimensional configurations. The unregularized law is recovered formally in the ideal limit $\tau \rightarrow 0$ whenever C_Ω remains sufficiently nonzero.

$$\lim_{\tau \rightarrow 0} R_{\{\Omega, \tau\}}[A] = \text{Norm}(P(-\Omega^2)P), \quad \text{when } \|P(-\Omega^2)P\|_F > 0 \quad (15)$$

The corresponding self-consistency functional is:

$$E_{\{\text{RRPG}, \tau\}}(A) = \|A - R_{\{\Omega, \tau\}}[A]\|_F^2 \quad (16)$$

This functional measures the mismatch between the current relational substrate and the covariant curvature target generated by its own projector geometry.

5. Local variational dynamics

The autonomous RRPG update is a finite-step relaxation toward the regularized target:

$$A_{t+1} = \text{Norm}[(1-\lambda)A_t + \lambda R_{\{\Omega, \tau\}}[A_t]] \quad (17)$$

The associated update direction is:

$$D_{\{\Omega, \tau\}}(A) = R_{\{\Omega, \tau\}}[A] - A \quad (18)$$

Finite-difference tests support $D_{\{\Omega, \tau\}}$ as a local descent direction for $E_{\text{RRPG}, \tau}$. Therefore the dynamics is best described as local variational self-consistency relaxation, not as direct maximization of curvature power and not as a proven global contraction map.

$$D E_{\{\text{RRPG}, \tau\}}(A)[D_{\{\Omega, \tau\}}] < 0 \quad (19)$$

6. Covariance and basis dependence

Under an orthogonal change of basis Q , the relational substrate and constraint directions transform as:

$$A \rightarrow Q A Q^T, \quad C_i \rightarrow Q C_i Q^T \quad (20)$$

The intrinsic objects transform covariantly:

$$P \rightarrow Q P Q^T, \quad \Omega \rightarrow Q \Omega Q^T, \quad C_\Omega \rightarrow Q C_\Omega Q^T, \quad R_{\{\Omega, \tau\}} \rightarrow Q R_{\{\Omega, \tau\}} Q^T \quad (21)$$

The relaxation energy is invariant:

$$E_{\{\text{RRPG}, \tau\}}(Q A Q^T) = E_{\{\text{RRPG}, \tau\}}(A) \quad (22)$$

By contrast, node-basis diagnostics such as $\text{diag}(P)$, K_i , S_r , and correlation islands are meaningful only relative to a chosen node representation. They are useful for interpretation and experiments but should not be presented as coordinate-free physical invariants.

7. Numerical certification audit

An integrated certification audit tested the paper-critical claims across matrix sizes (16,3), (20,4), (24,4), and (28,5), using $\tau = 10^{-5}$ and $\lambda = 0.04$. The audit included covariance checks, basis-invariance checks, finite-difference variational descent, and perturbation-based coarse stability tests.

Certified quantity	Result	Interpretation
Maximum covariance error	2.207e-11	Projector-curvature core and regularized target are basis-covariant.
Energy basis difference	3.491e-12	$E_{\text{RRPG}, \tau}$ is invariant under orthogonal basis changes.
Variational descent rate	1.000	$D_{\{\Omega, \tau\}}$ was a descent direction in all tested cases.
Beats-random rate	1.000	The update direction outperformed random symmetric directions.
Descent margin	1.143	Mean advantage over random directions.
Macro-stability, regularized	0.945	Coarse projector-curvature observables remain stable.
Macro-stability, null	0.887	Regularized dynamics outperformed null dynamics.

Table 1. Integrated RRPG certification results.

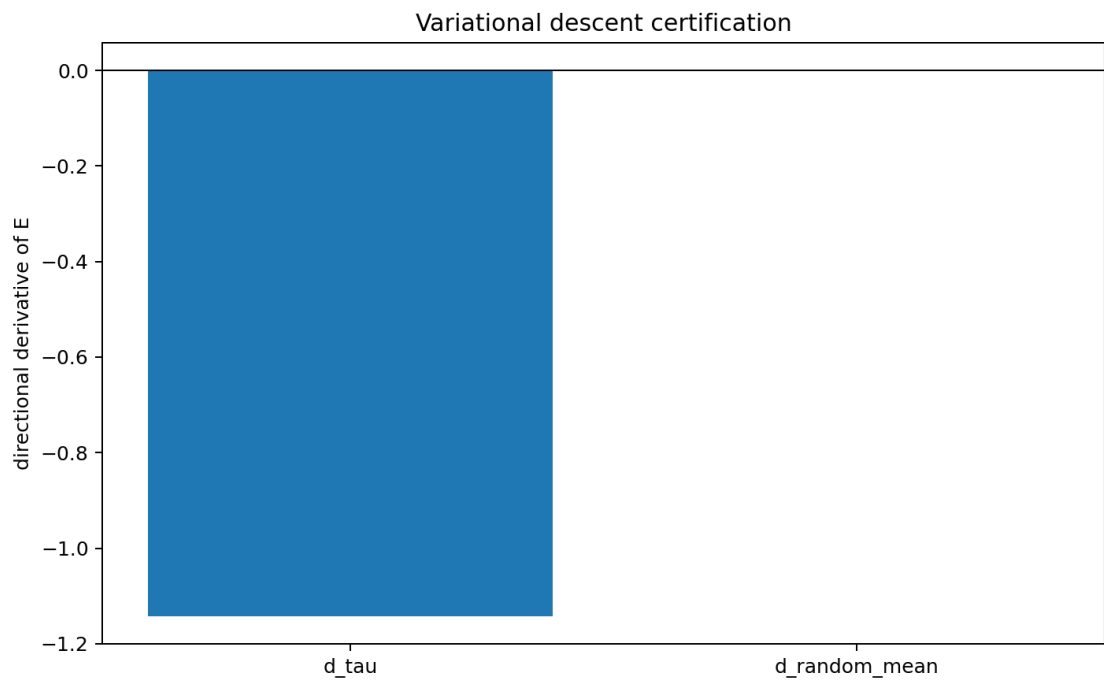


Figure 1. Finite-difference certification of variational descent.

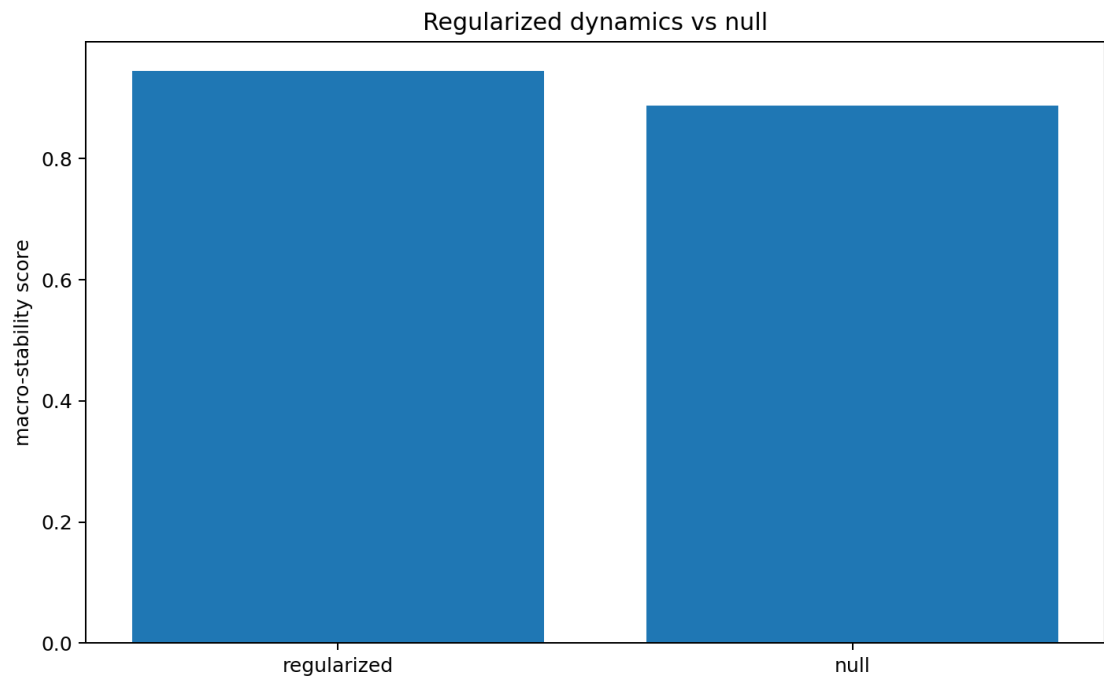


Figure 2. Macro-stability comparison between regularized dynamics and null dynamics.

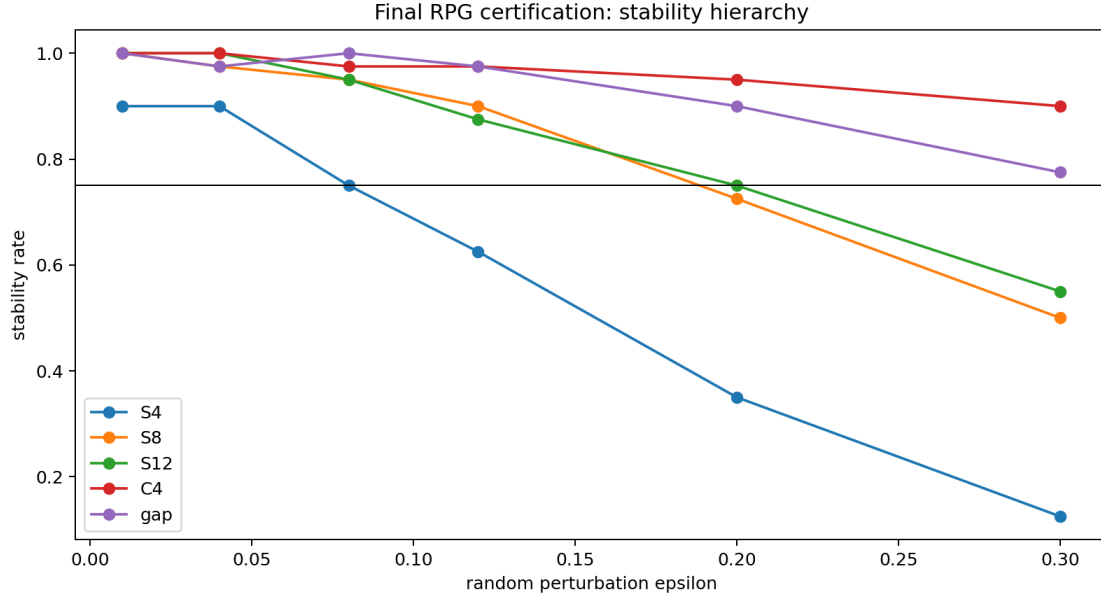


Figure 3. Stability hierarchy for random perturbations.

7.1 Stability hierarchy

The audit found that exact node-sector identity is the most fragile diagnostic, while coarse curvature and gap structure are substantially more robust. This supports a resolution hierarchy for RRPG observables.

Perturbation type	S4 epsilon_crit	S8 epsilon_crit	C4 epsilon_crit	Gap epsilon_crit
Random	0.08	0.12	0.3	0.3
Sector-structured	0.01	not central	0.3	0.2

Table 2. Critical perturbation radii for selected observables.

The stability hierarchy is therefore: exact node sectors S4 are fragile; broader sectors S8 and S12 are more stable; core enrichment C4 and the spectral gap Delta are the most robust tested geometric diagnostics.

8. Interpretation and physical meaning

RRPG should be interpreted as a finite relational geometry framework. Its primary geometric object is not a coordinate point, but a projector-selected subspace. Local curvature is not imposed externally, but appears as the obstruction to commuting projector deformations. The regularized dynamics says that a relational state tends locally toward consistency with the curvature-power structure generated by its own projector geometry.

This does not yet make RRPG a complete physical theory of spacetime. Instead, it gives a mathematically precise mechanism by which geometric-like structure can be defined and stabilized without assuming a pre-existing metric manifold. Potential applications include pre-geometric modeling, relational network analysis, geometric diagnostics for learned representations, and controlled synthetic models of emergent curvature.

9. Limitations and open development directions

The current framework has clear limits. First, the dynamics is locally variational but not proven to be a global contraction to a unique fixed point. Second, exact node-sector identity is resolution-dependent and fragile under strong perturbations. Third, the holonomy/transport layer is not yet validated as a robust dynamical law.

Future work should develop a stronger transport theory, possibly using discrete connection maps on the frame bundle rather than substrate-level relaxation. It should also test continuum limits, larger ensembles, non-symmetric substrates, stochastic perturbations, and links to known geometric structures such as Grassmannian geometry and operator-valued curvature.

10. Conclusion

Regularized Relational Projector Geometry provides a basis-covariant framework in which geometry is generated from spectral projectors over a relational substrate. The regularized curvature target $R_{\{\Omega, \tau\}}$ and the energy $E_{RRPG, \tau}$ define a local variational self-consistency dynamics. Numerical certification supports covariance, energy invariance, finite-difference descent, and coarse projector-curvature stability above null controls. The strongest current claim is local regularized variational geometry, not global attractor dynamics. RRPG therefore offers a disciplined mathematical foundation for projector-defined emergent geometry while leaving full holonomy and transport dynamics as the central open extension.

Appendix A. Algorithmic specification

1. Normalize a symmetric relational substrate A .
2. Choose rank k and constraint directions C_1, C_2 .
3. Construct the constrained substrate $A(a,b)$.
4. Compute the spectral projector $P(A)$.
5. Estimate projector derivatives by centered finite differences.
6. Compute $\Omega = P[\partial_1 P, \partial_2 P]P$.
7. Compute $C_\Omega = P(-\Omega^2)P$.
8. Compute $R_{\{\Omega, \tau\}} = \text{Norm}(C_\Omega + \tau A)$.
9. Update A by $A_{\text{next}} = \text{Norm}((1-\lambda)A + \lambda R_{\{\Omega, \tau\}})$.
10. Evaluate $E_{\text{RRPG}, \tau}$, gap , C_r , S_r , and projector distance diagnostics.

Appendix B. Paper-safe claims

Type	Statement
Safe claim	RRPG defines intrinsic geometry through spectral projectors and curvature obstruction.
Safe claim	The regularized target $R_{\{\Omega, \tau\}}$ is basis-covariant in the tested finite-dimensional setting.
Safe claim	$E_{\text{RRPG}, \tau}$ is basis-invariant and supports local variational descent.
Safe claim	Regularized dynamics preserves coarse projector-curvature geometry better than null dynamics.
Limitation	Exact node sectors are diagnostic and resolution-dependent, not invariant geometric primitives.
Limitation	Global basin contraction and complete holonomy/transport dynamics are not established.

Appendix C. Notation summary

Symbol	Meaning
A	normalized symmetric relational substrate
C_1, C_2	constraint/deformation directions
P	rank- k spectral projector
d_P	projector distance $\ P(A) - P(B)\ _F$
Ω	curvature obstruction $P[\partial_1 P, \partial_2 P]P$
K_i	node-basis curvature profile
S_r	top- r diagnostic sector
C_r	sector curvature enrichment
C_Ω	intrinsic curvature-power operator $P(-\Omega^2)P$
$R_{\{\Omega, \tau\}}$	regularized covariant target
$E_{\text{RRPG}, \tau}$	self-consistency energy
λ	relaxation step size
τ	regularization parameter